

OPTICS

EXPERIMENT E-15

LAWS OF REFLECTION

DATE: 14TH NOVEMBER, 2022.

NAME: SHISAMA SHIVKA AITON.

REG. NO: FIG/1978/2022

PARTNERS: OMBUI JOSHUA - FIG/1983/2022

YVONE ATIENO - FIG/1976/2022

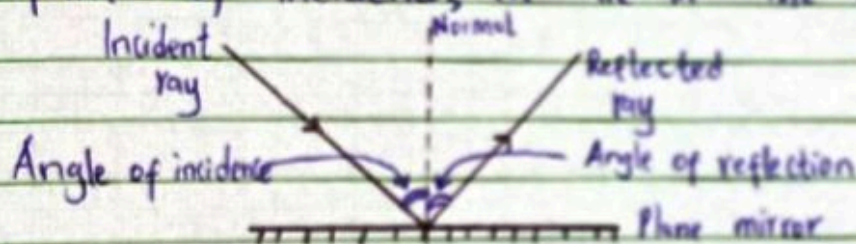
OBJECTIVES

1. To verify the laws of reflection.
2. To investigate some properties of spherical mirrors.

INTRODUCTION

The laws of reflection ~~are~~ for plane mirrors are:

- (i) The angle of incidence is equal to the angle of reflection.
- (ii) The incident ray, the reflected ray and the normal, at the point of incidence, all lie in the same plane.



For spherical mirrors, the position of an image formed ~~by~~ can be obtained either by ~~geometrical~~ geometrical construction or by using general relationship for lenses and mirrors given by;

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f} \quad \text{where } \begin{array}{l} v = \text{image distance} \\ u = \text{object distance} \\ f = \text{focal length} \end{array}$$

In geometrical optics, the following rules apply:

- (i) A ^(ray 1) ray from the object entering the mirror parallel to the principal axis is reflected back through the focus.
- (ii) A ^(ray 2) ray from the object entering the mirror through the center of curvature (C) is reflected back along the same path.

(iii) A ray (ray 3) from the object entering the mirror through the focus emerges parallel to the principal axis.

(a) Principle rays of a converging mirror

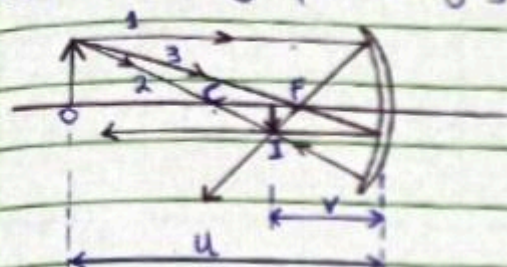


Image is real but inverted.

(b) Principle rays for a diverging mirror

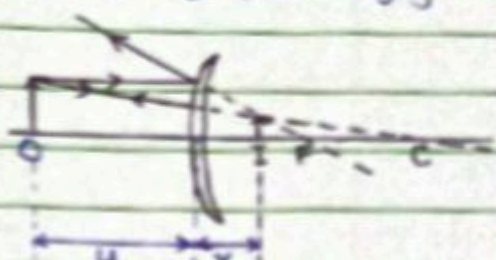


Image is upright but virtual.

APPARATUS

Plane mirror.

Pins.

Drawing board.

Paper.

Optical bench and fittings

A lamp-house (illuminated object)

Concave and convex mirrors.

Screen.

METHOD

Part A: Plane Mirror

The mirror was placed along line BC drawn on the paper placed on the drawing board.

Two pins, P_1 and P_2 , were inserted into the drawing board. The directions of Pins P_3 and P_4 was observed until they were in the same line as pins P_1 and P_2 .

A line was drawn on the drawing paper marking the reflecting surface of the mirror.

The mirror and pins were removed from the drawing board. Line AN (the normal) was drawn perpendicular to BC.

Points P_1, P_2 and P_3, P_4 were joined and extended the lines to point A.

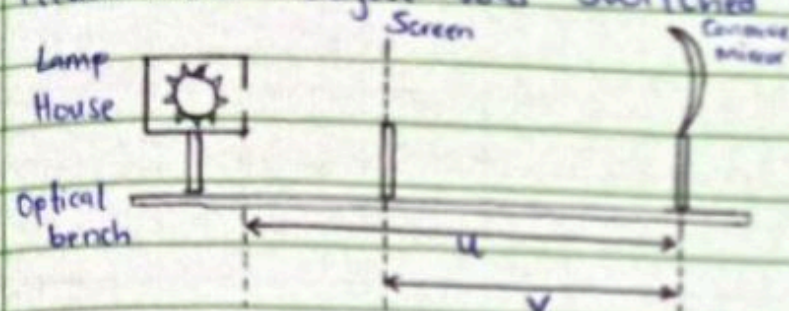
The angles of incidence and reflection were measured and recorded in the worksheet 15-1.

The original ray diagram was—

Part B: Spherical Mirrors

(a) Concave Mirrors

The apparatus were set up as shown below and the illuminated object was switched on.



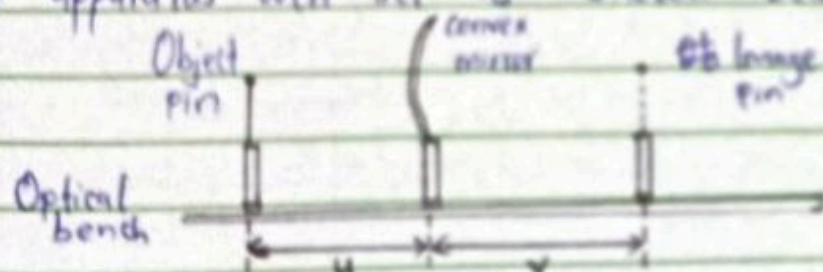
The concave mirror was moved slowly away from the illuminated object until it projects an image of the illuminated object on itself. The mirror was moved so that a good clear focus is obtained. The distances between the mirror and object (u) / image (v) were measured and recorded. $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$ Half the distance was the focal length, f .

The mirror was set at a distance greater than $2f$ away from the object. The screen was then used to locate the image and the image distance was measured for $u > 2f$, $u = 2f$, $2f > u > f$, $u = f$ and $u < f$.

The results were recorded in worksheet 15-2 (i). The types of images observed in each case was also recorded.

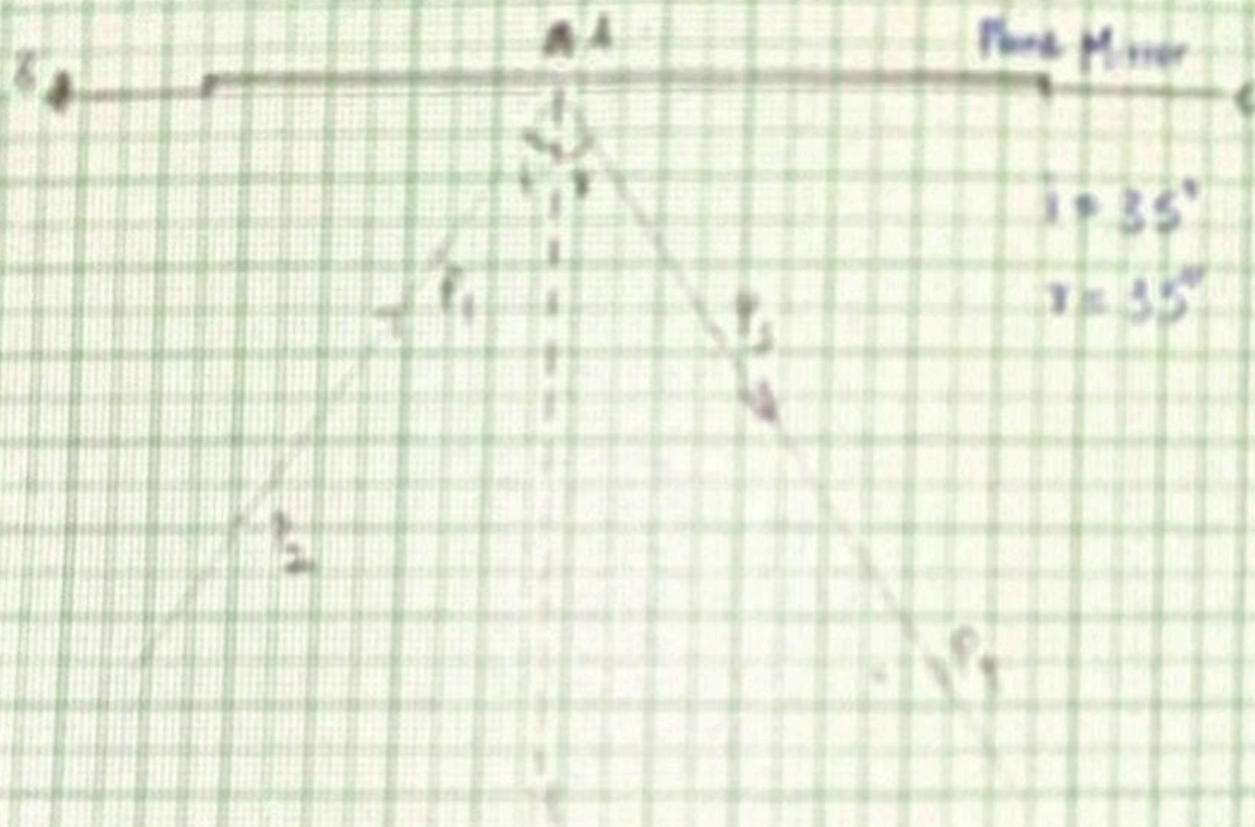
(b) Convex Mirrors

The apparatus were set as shown below with one pin as object.



The other pin was used as a search pin to locate the virtual image in the convex mirror using the method of no parallax. This was repeated twice and the results recorded in worksheet 15-2 (ii).

PART A: PLANE MIRROR



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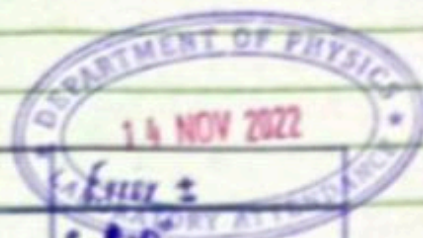
REG NO: F16/1978/2022

PARTNERS: OMBVI JOSHUA - F16/1983/2022

YVONE ATIENO - F16/1976/2022

WORKSHEET 15-1: PLANE MIRROR

	Angle	$\angle \text{incidence} =$
Angle of incidence, i	35°	0°
Angle of reflection, r	35°	0°



WORKSHEET 15-2: SPHERICAL MIRRORS

(i) Concave Mirror

Object distance, $u = 32.5 \pm 0.1 \text{ cm}$

Image distance, $v = 32.5 \pm 0.1 \text{ cm}$

Focal length, $f = \frac{1}{2}u = 16.25 \pm 0.01 \text{ cm}$

	$u \pm 0.1 \text{ cm}$	$(\text{cm}) \frac{1}{u} \pm 0.0001$	$(\text{cm}) v \pm 0.1$	$\frac{1}{v} \pm 0.0001$	Is image		
		± 0.0001		± 0.0001	real	upright	Magnified
$u > 2f$	46.0	0.0217	22.0	0.0455	✓	x	✓
$u = 2f$	32.5	0.0308	32.5	0.0308	✓	x	✓
$2f > u > f$	24.0	0.0417	44.0	0.0227	✓	x	✓
$u = f$	16.2	0.0617	—	—	—	—	—
$u < f$	12.0	0.0833	-48.0	-0.0208	x	x	x

(ii) Convex Mirror	$u \pm 0.1 \text{ cm}$	$v \pm 0.1 \text{ cm}$	$f \text{ cm}$
	23.0	55.0	16.2
	29.0	37.0	16.3
	33.0	32.0	16.4
		Average f	16.3

Graph of k_u against k_v of concrete masonry



ANALYSIS AND CALCULATIONS

For concave mirror, a graph of $1/v$ against $1/u$ was plotted and the value of f was determined from it.

The value of f is

When $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

When $1/u = 0,$
 $\frac{1}{f} = \frac{1}{v} = 0.0540$

Thus $f_1 = v$
 $= 18.5$

When $1/v = 0,$
 $\frac{1}{f} = \frac{1}{u} = 0.0940$

Thus $f_2 = u = 13.5$

The average $f = \frac{f_1 + f_2}{2} = \frac{18.5 + 13.5}{2} = \frac{32}{2}$
 $= 16.0 \text{ cm}$

~~For plane mirror, refractive index $n = \sin$~~

For convex mirror, the error in f was calculated as below.

The sort of image produced was virtual, upright and diminished.

$$\frac{f_1 + f_2 + f_3}{3} = \frac{0.0001 + 0.0001 + 0.0001}{3}$$

$$= \frac{0.0003}{3} = \underline{\underline{0.0001 \text{ cm}}}$$

DISCUSSIONS

In the case of plane mirror, the laws of reflection were observed because angle of incidence = angle of reflection = 35° .

Also the incident ray, reflected ray and the normal, all lie ~~at~~ on the same plane at the point of incidence as in the ray diagram.

From the graph of $1/v$ against $1/u$ in the case of concave mirror, the focal length was found to be 16.0 cm which compares with the estimated one ~~to~~ obtained from focusing the illuminated object on itself, 16.2 cm.

The ~~for~~ type of image produced by concave mirror was determined by the distance the object is from the mirror.

The focal length of the ~~convex~~ mirror was found to be 16.3 cm.

The image formed was of one type; virtual, diminished and upright.

CONCLUSIONS

From the experiment, the laws of reflection were verified by the plane mirror.

Some of the properties of spherical mirrors were investigated and include the type of images formed by concave mirror (mostly real, inverted and magnified) and convex mirror (virtual, upright and diminished).

REFERENCES

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www.geeksforgeeks.org > laws-of-reflection
www.physics-louisville.edu > lo-s-mirror